

# Monte Carlo Pricing of European Options: Variance Reduction via Antithetic Variates and Validation Against the Black-Scholes Model

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## Abstract

This paper presents a Monte Carlo engine for pricing European call and put options under the risk-neutral Geometric Brownian Motion (GBM) measure. The estimator is built with an antithetic variates variance-reduction scheme and is systematically validated against the closed-form Black-Scholes formula. Using 100,000 simulated antithetic paths, the Monte Carlo price for an at-the-money call and put converges to within statistical noise of the analytical benchmark. A convergence study over simulation counts ranging from  $10^2$  to  $10^5$  confirms the expected  $O(1/\sqrt{N})$  reduction in pricing error, and a dedicated unit-test suite further validates put-call parity, boundary behaviour, and the variance-reduction property of the antithetic scheme.

## 1 Introduction

The Black-Scholes formula provides an elegant closed-form solution for European vanilla options, but it relies on restrictive assumptions (constant volatility, continuous trading, no transaction costs) that break down for most exotic or path-dependent payoffs encountered in practice. Monte Carlo simulation offers a flexible alternative capable of pricing virtually any payoff structure, at the cost of statistical (rather than exact) accuracy. Before a Monte Carlo engine can be trusted on instruments with no analytical solution, it must first be rigorously validated on the simplest case for which an exact answer is known. This project builds such an engine for vanilla European options, benchmarks it against Black-Scholes, and quantifies how quickly the simulation error vanishes as the number of paths grows.

## 2 Theoretical Framework

### 2.1 Risk-Neutral Asset Dynamics

Under the risk-neutral measure, the underlying asset price is assumed to follow Geometric Brownian Motion. The terminal price  $S_T$  admits the closed-form solution:

$$S_T = S_0 \exp \left[ \left( r - \frac{1}{2} \sigma^2 \right) T + \sigma \sqrt{T} Z \right], \quad Z \sim \mathcal{N}(0, 1) \quad (1)$$

where  $S_0$  is the spot price,  $r$  the continuously-compounded risk-free rate,  $\sigma$  the annualised volatility,  $T$  the time to maturity, and  $Z$  a standard normal random variable. This formulation allows the terminal distribution to be sampled directly, without discretising the path, since the GBM solution is known in closed form at any fixed horizon.

### 2.2 Monte Carlo Estimator, Variance Reduction, and the Black-Scholes Benchmark

The discounted expected payoff defines the option price:

$$V_0 = e^{-rT} \mathbb{E}[f(S_T)], \quad f(S_T) = \begin{cases} \max(S_T - K, 0) & \text{(call)} \\ \max(K - S_T, 0) & \text{(put)} \end{cases} \quad (2)$$

To reduce the variance of the Monte Carlo estimator without additional random draws, the engine uses *antithetic variates*: for every draw  $Z_i$ , the mirrored draw  $-Z_i$  is also evaluated, and the two payoffs are averaged,

$$\hat{f}_i = \frac{f(S_T(Z_i)) + f(S_T(-Z_i))}{2}. \quad (3)$$

Because  $f(Z)$  and  $f(-Z)$  are negatively correlated for monotone payoffs such as call and put values,  $\hat{f}_i$  has strictly lower variance than a single unpaired draw, at no extra simulation cost.

The Monte Carlo price is then validated against the closed-form Black-Scholes price,

$$C = S_0 N(d_1) - K e^{-rT} N(d_2), \quad d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2} \sigma^2) T}{\sigma \sqrt{T}}, \quad d_2 = d_1 - \sigma \sqrt{T}, \quad (4)$$

with the put price recovered by put-call parity,  $P = C - S_0 + K e^{-rT}$ .

## 3 Implementation Methodology

### 3.1 Software Architecture

The engine is organised into three components. An `EuropeanOption` dataclass encapsulates the contract parameters ( $S$ ,  $K$ ,  $T$ ,  $r$ ,  $\sigma$ , option type) and enforces validity at construction time, rejecting non-positive spot prices, non-positive maturities, non-positive volatilities, or option types other than 'call'/'put'. A `MonteCarloEngine` class consumes an `EuropeanOption` and returns, via its `price()` method, a dictionary containing the price estimate, its standard error, and the associated 95% confidence interval; the antithetic scheme can be toggled off to isolate its variance-reduction effect. A `BlackScholesModel` class implements the closed-form formula of Section 2.2 and serves as the ground-truth benchmark for both the point estimate and the convergence study.

### 3.2 Convergence and Validation Methodology

To produce a smooth convergence curve without re-seeding noise between runs, a single batch of  $N_{\max} = 100,000$  standard normal draws is generated once (seed fixed at 42), and both the direct and antithetic terminal prices are computed from it. The running average of the discounted payoff is then evaluated at 80 geometrically-spaced simulation counts between  $10^2$  and  $10^5$ , together with its running standard error, which traces the convergence of the Monte Carlo price toward the Black-Scholes benchmark and the associated shrinkage of the confidence band.

Correctness is further enforced by an automated test suite of nine unit tests, covering: exact put-call parity in the analytical formula; positivity of at-the-money option time value; convergence of deep in-the-money and deep out-of-the-money prices to their respective limiting cases; agreement between the Monte Carlo and Black-Scholes prices within two standard errors; put-call parity holding in the Monte Carlo estimator; the strict variance reduction achieved by antithetic variates relative to naive sampling; and rejection of invalid option parameters. All nine tests pass.

## 4 Results and Discussion

### 4.1 Pricing Accuracy and Statistical Validation

The engine was run for an at-the-money option ( $S_0 = 100$ ,  $K = 100$ ,  $T = 1$  year,  $r = 5\%$ ,  $\sigma = 20\%$ ) with 100,000 antithetic Monte Carlo paths (seed 42). Table 1 summarises the comparison against the analytical Black-Scholes price.

Table 1: Monte Carlo vs. Black-Scholes pricing comparison ( $N = 100,000$ , antithetic variates)

| Option | MC Price | Std. Error | 95% CI             | BS Price | MC – BS |
|--------|----------|------------|--------------------|----------|---------|
| Call   | 10.4763  | 0.0234     | [10.4303, 10.5222] | 10.4506  | 0.0257  |
| Put    | 5.5842   | 0.0149     | [5.5550, 5.6134]   | 5.5735   | 0.0107  |

In both cases the Black-Scholes price lies inside the 95% confidence interval produced by the Monte Carlo estimator, and the absolute pricing error is on the order of  $10^{-2}$ , consistent with the residual statistical noise expected at  $N = 10^5$  paths. Combined with the full pass of the nine-test validation suite, these results confirm that the engine reproduces the analytical benchmark to within sampling error and is therefore suitable as a foundation for pricing payoffs that lack a closed-form solution.

## 4.2 Visualization

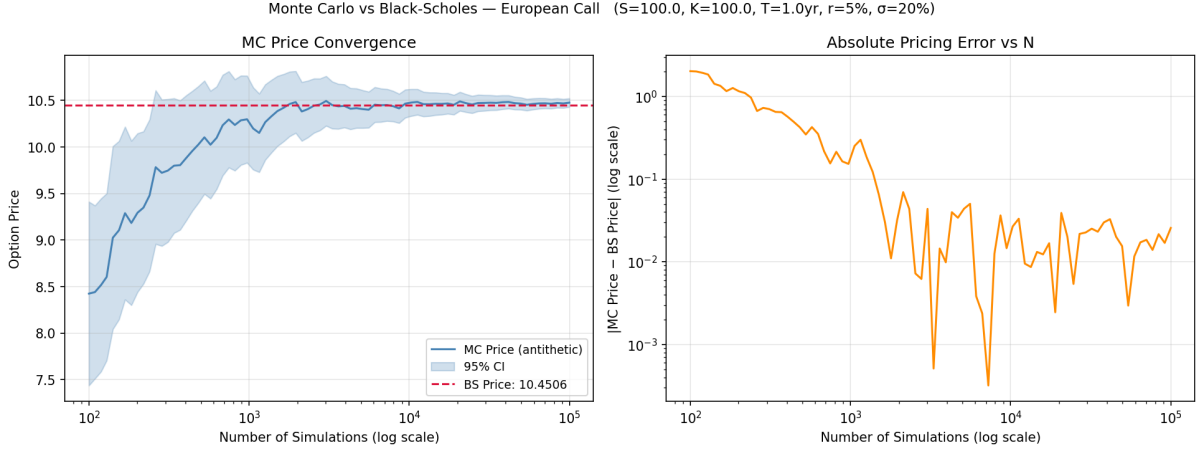


Figure 1: Monte Carlo convergence for the at-the-money European call ( $S_0 = 100$ ,  $K = 100$ ,  $T = 1\text{yr}$ ,  $r = 5\%$ ,  $\sigma = 20\%$ ). Left: running Monte Carlo price (antithetic variates) with its 95% confidence band, converging to the Black-Scholes price (dashed line) as  $N$  grows on a log scale. Right: absolute pricing error  $|\text{MC} - \text{BS}|$  on a log-log scale, decaying at the theoretically expected rate of  $O(1/\sqrt{N})$ .

## 5 Conclusion

This project demonstrates that a Monte Carlo engine built on risk-neutral Geometric Brownian Motion, sharpened with antithetic variates, reproduces the Black-Scholes price for European options to within statistical noise after 100,000 simulated paths, with the pricing error decaying at the theoretically predicted  $O(1/\sqrt{N})$  rate. Beyond validating the simulation methodology itself, this benchmark establishes a tested and trustworthy foundation: the same engine architecture extends naturally to path-dependent or early-exercise payoffs for which Black-Scholes offers no closed-form solution, and for which variance-reduction techniques such as antithetic variates become essential to keep simulation costs tractable.

## References

- [1] Black, F., & Scholes, M. (1973). *The Pricing of Options and Corporate Liabilities*. Journal of Political Economy, 81(3), 637-654.
- [2] Boyle, P. P. (1977). *Options: A Monte Carlo Approach*. Journal of Financial Economics, 4(3), 323-338.
- [3] Glasserman, P. (2004). *Monte Carlo Methods in Financial Engineering*. Springer.
- [4] Hull, J. C. (2018). *Options, Futures, and Other Derivatives*. Pearson.